

Definition. $\lim_{x \rightarrow c} f(x) = L$ iff for every $\varepsilon > 0$, we can find a corresponding $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - c| < \delta$

- Example. Suppose $f(x) = 4x - 1$. Apply the $\varepsilon - \delta$ definition of a limit to show that $\lim_{x \rightarrow 3} f(x) = 11$. Hint: Starting with

$$|f(x) - 11| < \varepsilon, \text{ try to establish inequality of the form } |x - 3| < \dots$$

cal. Given one $\varepsilon > 0$

$$|f(x) - 11| < \varepsilon$$

$$\Leftrightarrow |4x - 1 - 11| < \varepsilon$$

$$\Leftrightarrow |4x - 12| < \varepsilon$$

$$\Leftrightarrow |x - 3| < \frac{\varepsilon}{4}$$

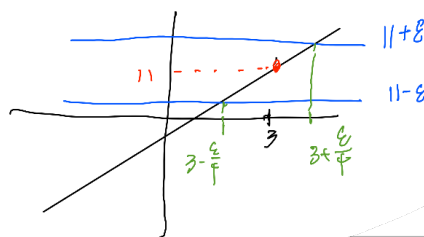
proof:

Given one $\varepsilon > 0$

we can find $\delta = \frac{\varepsilon}{4}$

such that $|f(x) - 11| < \varepsilon$ whenever

$$0 < |x - 3| < \delta$$

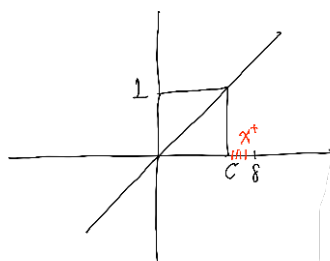
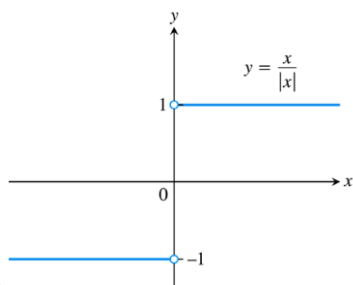


ch. 2.4

Definitions. We say $f(x)$ has a right-hand limit L at c , written as $\lim_{x \rightarrow c^+} f(x) = L$, if for every number $\varepsilon > 0$, there exists a corresponding $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } c < x < c + \delta.$$

- What's the definition of a left-hand limit?



- Example. Suppose $f(x) = \sqrt{x} - 1$. Prove that $\lim_{x \rightarrow 0^+} f(x) = -1$ by the $\varepsilon - \delta$ condition. Hint: Starting with $|f(x) - (-1)| = |\sqrt{x} - 1 - (-1)| < \varepsilon$, try to find some result of the form $0 < x - 0 < \dots$

Given one $\varepsilon > 0$

$$\Leftrightarrow |\sqrt{x} - 1 - (-1)| < \varepsilon$$

$$\Leftrightarrow |\sqrt{x}| < \varepsilon$$

$$\Rightarrow \text{if } 0 < x < \varepsilon^2$$

$$\text{then } |\sqrt{x}| < \varepsilon$$

leading to

$$|f(x) - (-1)| < \varepsilon$$

Given one $\varepsilon > 0$

we find corresponding $\delta = \varepsilon^2$

such that whenever $0 < x < \delta = \varepsilon^2$

